



CHAPTER FOUR

TIME-VARYING FIELD

NMJ21803 ELECTROMAGNETIC THEORY



BY
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SUB-CHAPTER

FARADAY'S LAW

LENZ'S LAW

TRANSFORMER

MOTIONAL ELECTROMOTIVE FORCES

DISPLACEMENT CURRENT

MAXWELL'S EQUATION



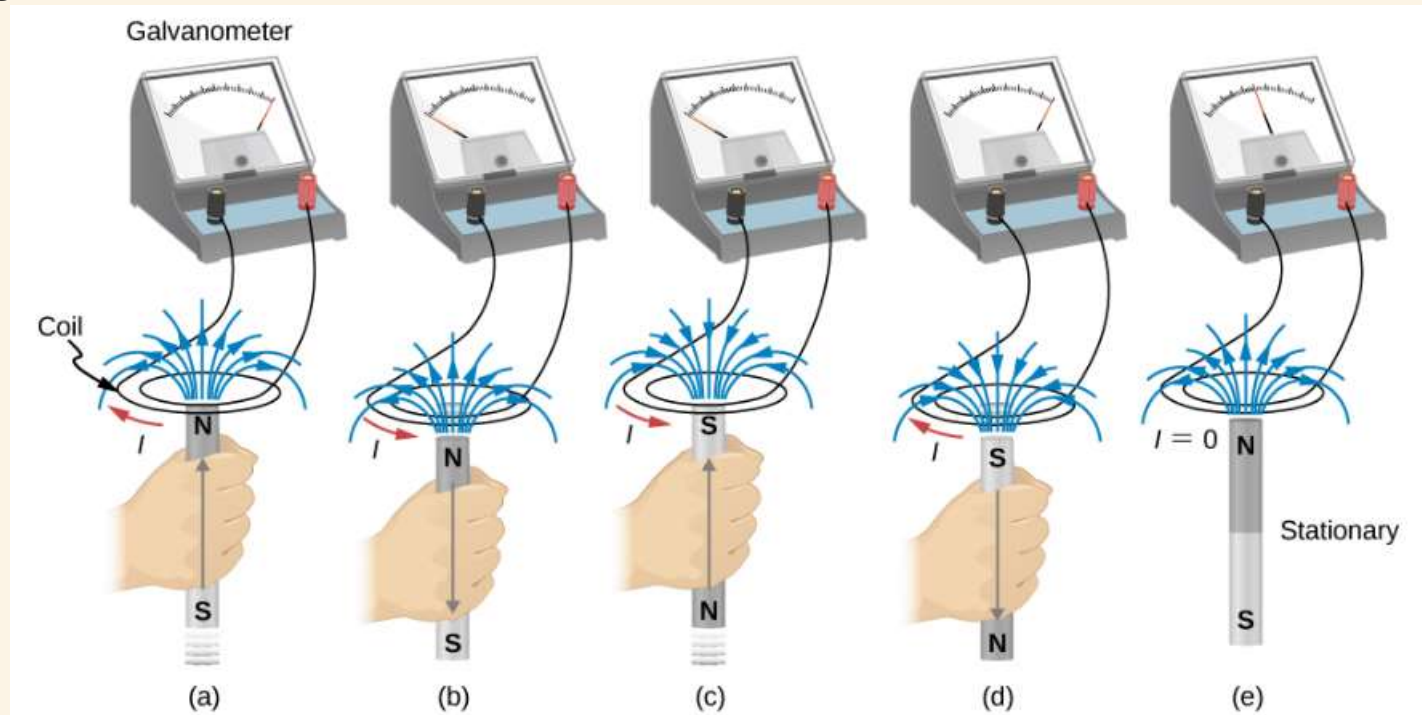
FARADAY'S LAW

By the end of this subtopic, you will be able to:

- ✓ Determine the magnetic flux through a surface, knowing the strength of the magnetic field, the surface area, and the angle between the normal to the surface and the magnetic field.
- ✓ Use Faraday's law to determine the magnitude of induced emf in a closed loop due to changing magnetic flux through the loop.
- ✓ Use Lenz's law to determine the direction of induced emf whenever a magnetic flux changes.
- ✓ Use Faraday's law with Lenz's law to determine the induced emf in a coil and in a solenoid.

FARADAY'S LAW

In 1831, Michael Faraday performed the first productive experiments concerning the effects of time-varying magnetic fields.



FARADAY'S LAW

An emf is induced when the magnetic field in the coil is **changed by pushing a bar magnet** into or out of the coil.

Emfs of **opposite signs** are produced by motion in **opposite directions**, and the directions of emfs are also reversed by reversing poles.

The **same results** are produced if the **coil is moved** rather than the magnet as it is the relative motion that is important.

The faster the motion, the greater the emf, and there is no emf when the magnet is stationary relative to the coil.

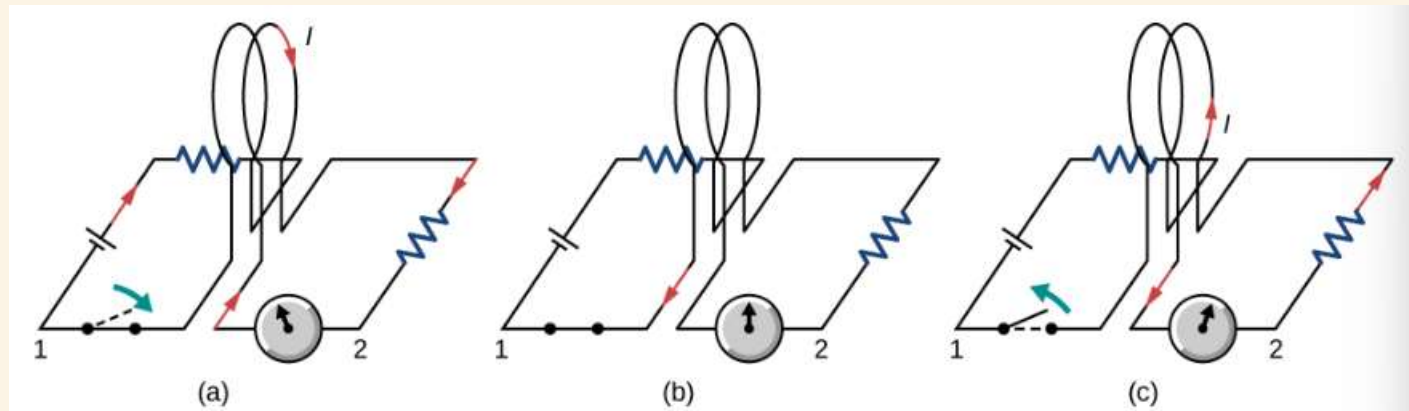
FARADAY'S LAW

A similar effect can be produced using two circuits, where a changing current in one circuit induces a current in a second, nearby circuit.

Closing the switch of circuit 1 produces a short-lived current surge in circuit 2.

If the switch remains closed, no current is observed in circuit 2.

Opening the switch again produces a short-lived current in circuit 2 but in the opposite direction from before.



FARADAY'S LAW

The emf induced is the negative change in the magnetic flux per unit time.

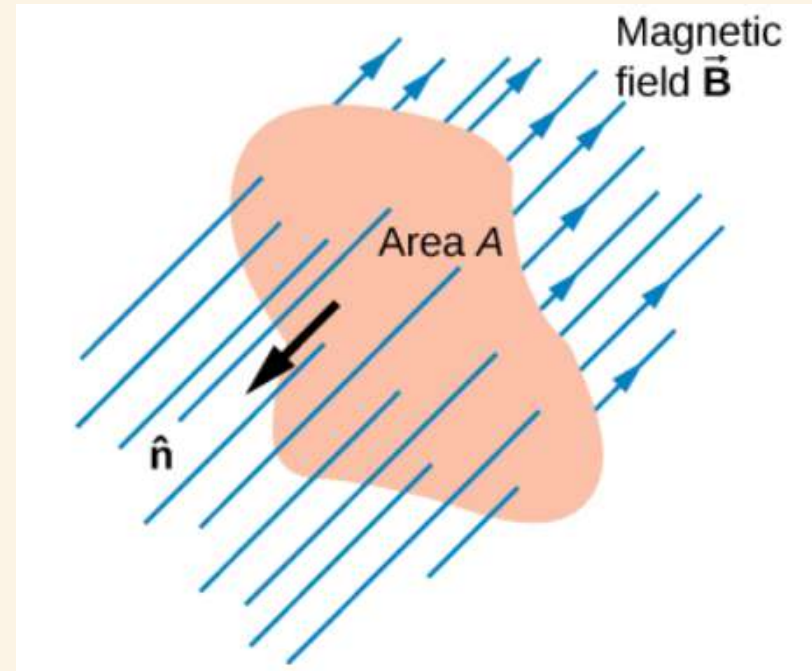
Any change in the magnetic field or change in orientation of the area of the coil with respect to the magnetic field will induce voltage (emf).

FARADAY'S LAW

The magnetic flux is a measurement of the amount of magnetic field lines through a given surface area.

If $\Phi_m = \int_S \vec{B} \cdot \hat{n} dA$, then the induced emf or voltage generated by a conductor or coil moving in a magnetic field is

$$\epsilon = -\frac{d}{dt} \int_S \vec{B} \cdot \hat{n} dA = -\frac{d\Phi_m}{dt}$$



FARADAY'S LAW

The SI unit for magnetic flux is the weber (Wb).

$$1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$$

The magnetic field unit is expressed as weber per square meter (Wb/m^2).

The net magnetic flux through the circuits is **N** times the flux through one turn, and Faraday's law is written as:

$$\epsilon = -\frac{d}{dt}(N\Phi_m) = -N\frac{d\Phi_m}{dt}$$

EXAMPLE

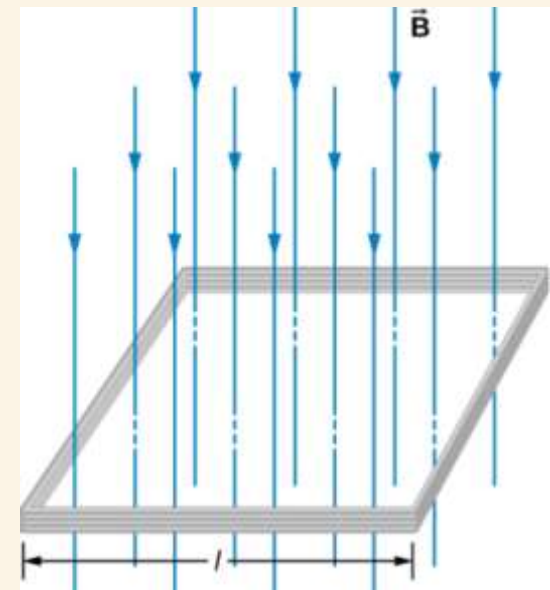
The square coil of Figure below has sides $l = 0.25\text{m}$ long and is tightly wound with $N = 200$ turns of wire. The resistance of the coil is $R = 5.0\ \Omega$. The coil is placed in a spatially uniform magnetic field that is directed perpendicular to the face of the coil and whose magnitude is decreasing at a rate $dB/dt = -0.04\ \text{T/s}$.

- (a) What is the magnitude of the emf induced in the coil?
- (b) What is the magnitude of the current circulating through the coil?

ANSWER:

(a) $0.5\ \text{V}$

(b) $0.1\ \text{A}$



SOLUTION

1. Find magnetic flux: $\Phi_m = BA$, where $A = l^2$, therefore;

$$\Phi_m = Bl^2.$$

2. Find induced emf (using Faraday's Law): $\epsilon = -N \frac{d\Phi_m}{dt}$, where $\Phi_m = Bl^2$, therefore;

$$\epsilon = -N \frac{dBl^2}{dt} = -Nl^2 \frac{dB}{dt}$$

3. Substitute all known values in ϵ .

4. Find induced current (using Ohm's Law): $V = IR$, therefore;

$$I = \frac{V}{R}$$

EXAMPLE

A closely wound coil has a radius of 4.0 cm, 50 turns, and a total resistance of 40 Ohm. At what rate must a magnetic field perpendicular to the face of the coil change in order to produce Joule heating in the coil at a rate of 2.0 mW?

ANSWER:

1.1 T/s

LENZ'S LAW

Lenz's law is stated as the direction of the induced emf drives current around a wire loop to always **oppose** the change in magnetic flux that causes the emf.

To determine an induced emf ϵ , find the magnetic flux Φ_m first, and then obtain $\frac{d\Phi_m}{dt}$. The magnitude of ϵ is given by:

$$\epsilon = \left| \frac{d\Phi_m}{dt} \right|$$

LENZ'S LAW

To use Lenz's law to determine the directions of induced magnetic fields, currents, and emfs:

1. Make a sketch of the situation for use in visualizing and recording directions.
2. Determine the direction of the applied magnetic field $\vec{B}$.
3. Determine whether its magnetic flux is increasing or decreasing.
4. Determine the direction of the induced magnetic field $\vec{B}$.

LENZ'S LAW

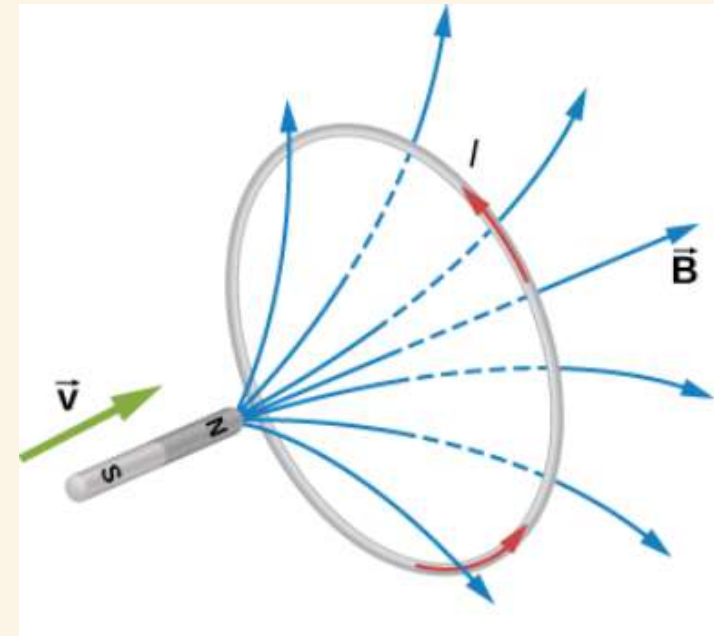
5. Use **right-hand rule** to determine the direction of the induced current I that is responsible for the induced magnetic field $\vec{B}$.

The direction of the induced emf can now drive a conventional current in this direction.

LENZ'S LAW

By RHR-2:

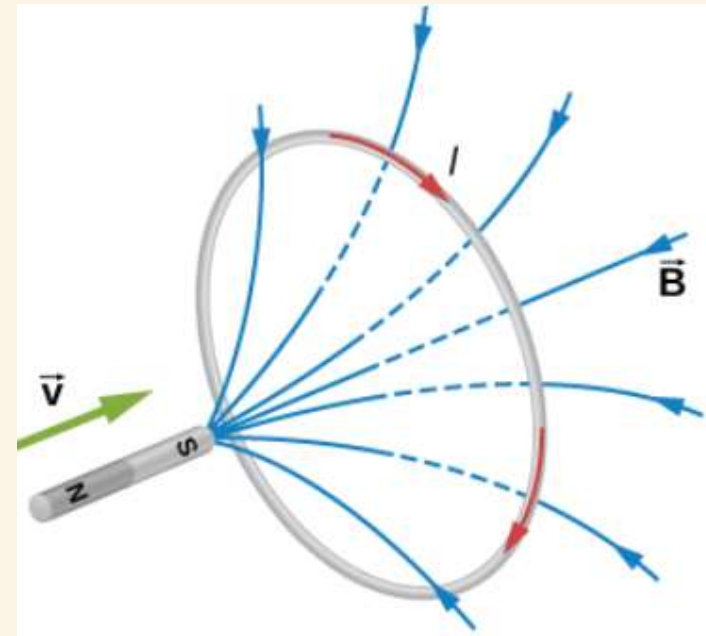
1. Place your thumb pointing against the magnetic field lines, which is toward the bar magnet.
2. Your fingers wrap in a counterclockwise direction as viewed from the bar magnet.



LENZ'S LAW

By RHR-2:

1. Place your thumb pointing against the magnetic field lines, which is away from the bar magnet.
2. Your fingers wrap in a clockwise direction, which is the direction of the induced current.



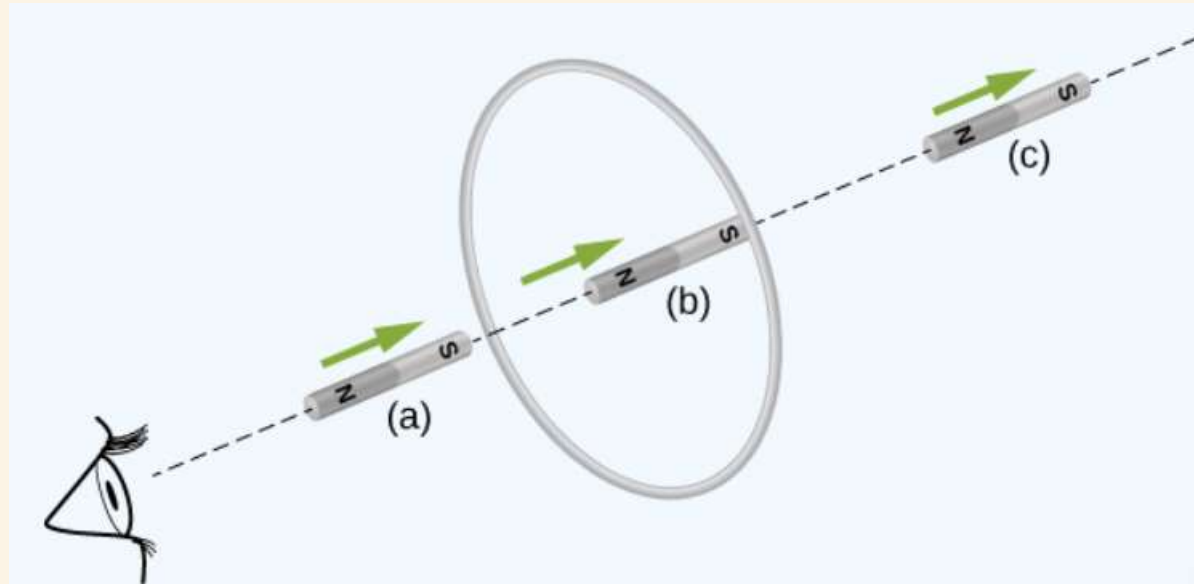
EXAMPLE

Find the direction of the induced current in the wire loop shown below as the magnet:

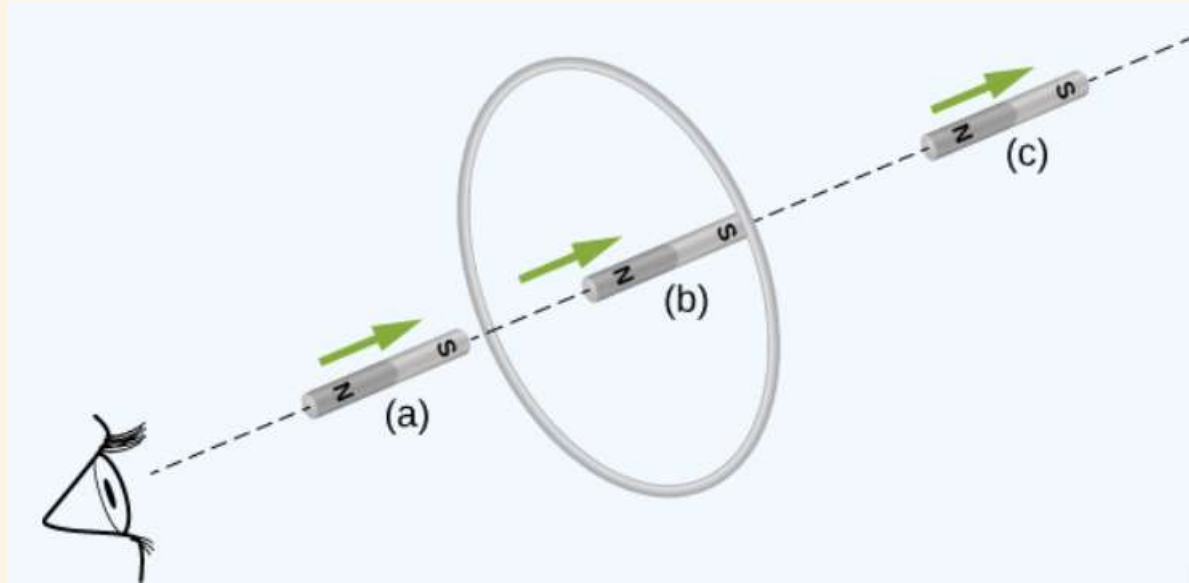
(a) enters,

(b) passes through and,

(c) leaves the loop.



EXAMPLE



(a) Enters: Clockwise

(b) Passes through: No current

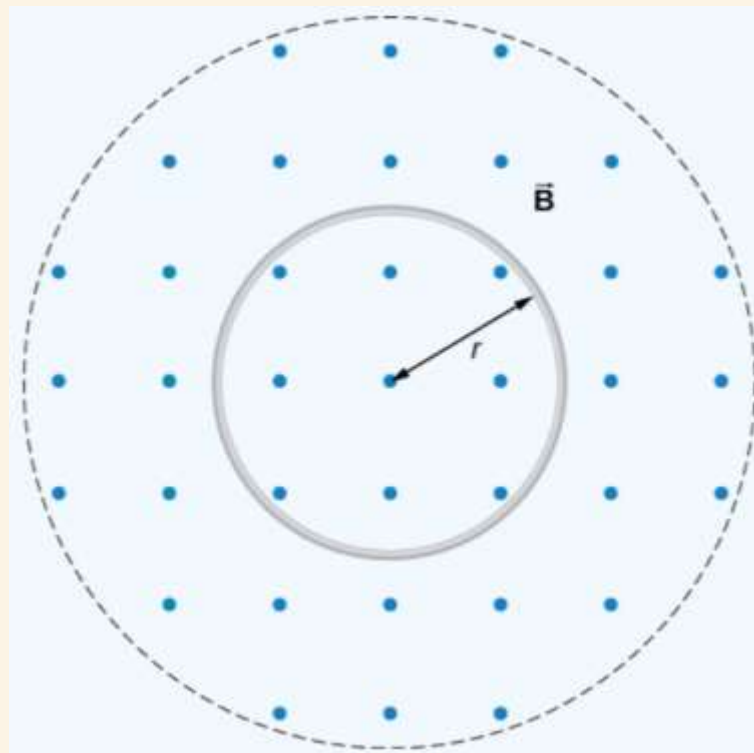
(c) leaves the loop: Counterclockwise

EXAMPLE

A magnetic field $\vec{B}$ is directed outward perpendicular to the plane of a circular coil of radius, $r = 0.5 \text{ m}$ (as in Figure in the next slide). The field is cylindrically symmetrical with respect to the center of the coil, and its magnitude decays exponentially according to $B = (1.5\text{T})e^{(5.0\text{s}^{-1})t}$, where $\mathbf{B}$ is in teslas and $\mathbf{t}$ is in seconds.

- (a) Calculate the emf induced in the coil at the times $t_1 = 0$, $t_2 = 5.0 \times 10^{-2}\text{s}$, $t_3 = 1.0\text{s}$.
- (b) Determine the current in the coil at these three times if its resistance is 10Ω .

EXAMPLE



EXAMPLE

SOLUTION:

Since $\vec{B}$ is perpendicular to the plane of the coil, the magnetic flux is given by:

$$\begin{aligned}\Phi_m &= B\pi r^2 \\ \Phi_m &= (1.5e^{(-5.0t)}T) \pi(0.5)^2 \\ \Phi_m &= 1.2e^{-(5.0s^{-1})t} \text{ Wb}\end{aligned}$$

From Faraday's law, the magnitude of the induced emf is:

$$\begin{aligned}\epsilon &= \left| \frac{d\Phi_m}{dt} \right| \\ \epsilon &= \left| \frac{d}{dt} (1.2e^{-(5.0s^{-1})t}) \right| \\ \epsilon &= 6.0e^{-(5.0s^{-1})t} \text{ V}\end{aligned}$$

EXAMPLE

SOLUTION:

For all three times, the sense of ϵ is counterclockwise; its magnitudes are:

$$\epsilon(t_1) = 6.0 \text{ V}$$

$$\epsilon(t_2) = 4.7 \text{ V}$$

$$\epsilon(t_3) = 40 \text{ V}$$

From Ohm's law, the respective currents are:

$$I(t_1) = \frac{\epsilon(t_1)}{R} = \frac{6.0}{10} = 0.6 \text{ A}$$

$$I(t_2) = \frac{\epsilon(t_2)}{R} = \frac{4.7}{10} = 0.47 \text{ A}$$

$$I(t_3) = \frac{\epsilon(t_3)}{R} = \frac{0.04}{10} = 4 \times 10^{-3} \text{ A}$$

EXAMPLE

The current through the windings of a solenoid with $n = 2000$ turns per meter is changing at a rate $di/dt = 3.0 \text{ A/s}$. The solenoid is 50 cm long and has a cross-sectional diameter of 3.0 cm. A small coil consisting of $N = 20$ closely wound turns wrapped in a circle of diameter 1.0 cm is placed in the middle of the solenoid such that the plane of the coil is perpendicular to the central axis of the solenoid. Assuming that the infinite-solenoid approximation is valid at the location of the small coil, determine the magnitude of the emf induced in the coil.

EXAMPLE

SOLUTION

Since the field of the solenoid is given by $B = \mu_0 n I$, the flux through each turn of the small coil is:

$$\Phi_m = \mu_0 n I \left(\frac{\pi d^2}{4} \right)$$

where d is the diameter of the coil.

EXAMPLE

SOLUTION

Now from Faraday's Law, the magnitude of the emf induced in the coil is:

$$\epsilon = \left| N \frac{d\Phi_m}{dt} \right|$$

$$\epsilon = \left| N \mu_0 n \frac{\pi d^2}{4} \frac{dI}{dt} \right|$$

$$\epsilon = 20(4\pi \times 10^{-7} T \cdot m/s)(2000 m^{-1}) \frac{\pi(0.01 m)^2}{4} (3.0 A/s)$$

$$\epsilon = 1.2 \times 10^{-5} V$$



THANK YOU

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MOTIONAL ELECTROMOTIVE FORCES

By the end of this subtopic, you will be able to:

- ✓ Determine the magnitude of an induced emf in a wire moving at a constant speed through magnetic field.
- ✓ .

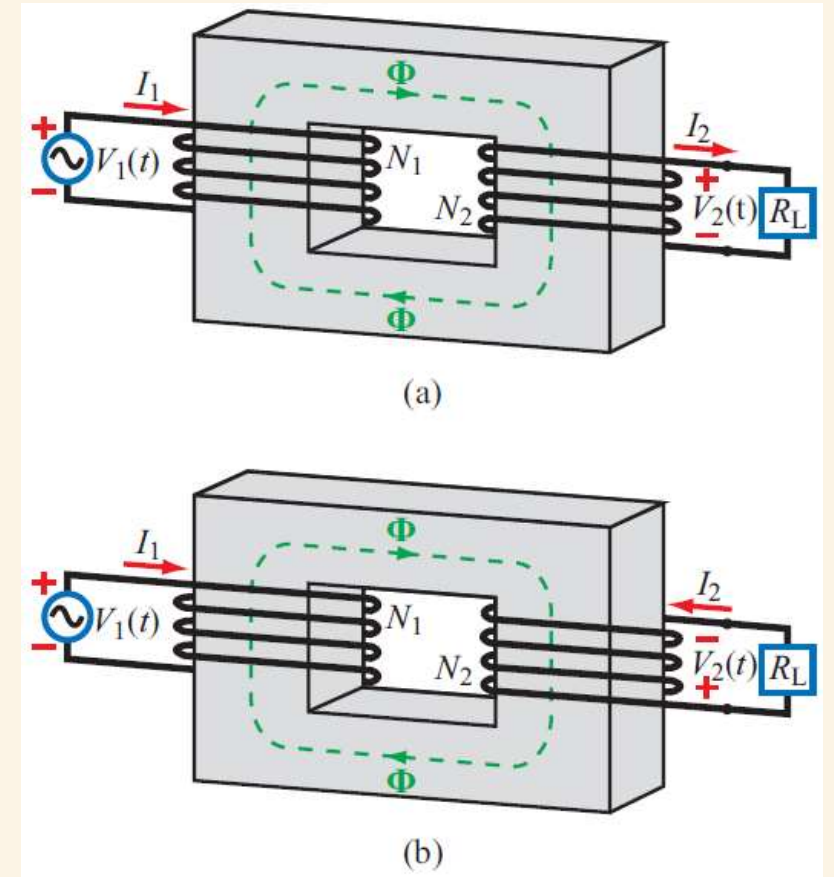
IDEAL TRANSFORMER

Transformer consist of 2 coils wound around a common magnetic core.

The primary coil has N_1 turns connected to a voltage source $V_1(t)$ and secondary coil has N_2 turns connected to R_L .

The directions of currents flowing in two coils: when I_1 and I_2 are both positive $\rightarrow$ flux generated by I_2 is opposite to that generated by I_1 .

Transformer $\rightarrow$ transfer currents, voltages and impedances between its primary and secondary circuits, and vice versa.



IDEAL TRANSFORMER

V_1 generates I_1 in primary coil $\rightarrow$ establishes flux Φ in magnetic core:

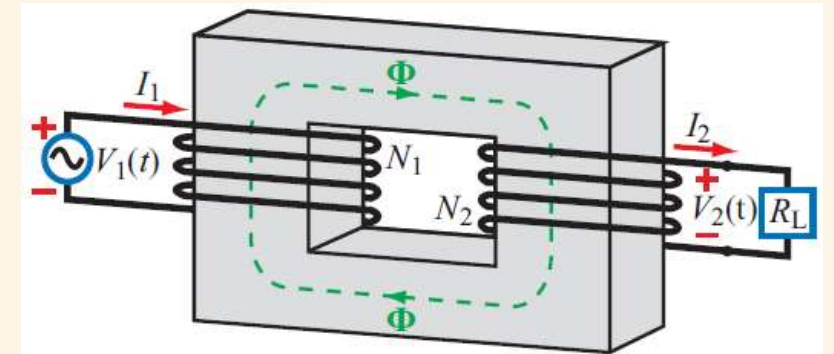
$$V_1 = -N_1 \frac{d\Phi}{dt}$$

Similar relation on secondary side:

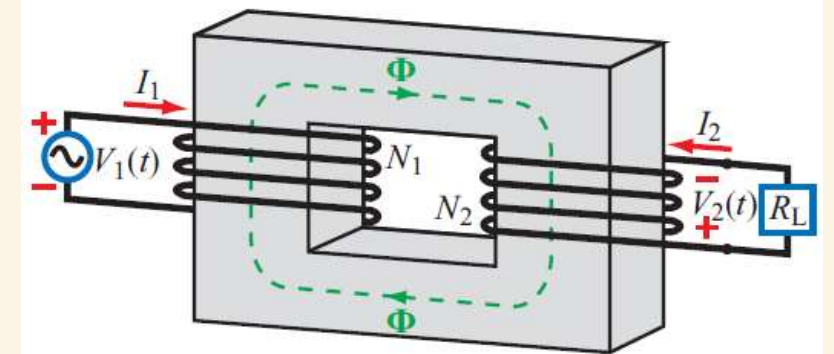
$$V_2 = -N_2 \frac{d\Phi}{dt}$$

Combination of both equations:

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$



(a)



(b)

IDEAL TRANSFORMER

In ideal lossless transformer:

$$P_1 = P_2 \rightarrow \frac{I_1}{I_2} = \frac{N_2}{N_1}$$

Equivalent input resistance:

$$R_{in} = \frac{V_1}{I_1} = \frac{V_2}{I_2} \left(\frac{N_1}{N_2} \right)^2 = \left(\frac{N_1}{N_2} \right)^2 R_L$$

In ideal lossless transformer:

$$P_1 = P_2 \rightarrow \frac{I_1}{I_2} = \frac{N_2}{N_1}$$

Equivalent input resistance:

$$R_{in} = \frac{V_1}{I_1} = \frac{V_2}{I_2} \left(\frac{N_1}{N_2} \right)^2 = \left(\frac{N_1}{N_2} \right)^2 R_L$$

EXAMPLE

A transformer on a utility pole steps the rms voltage down from 12 kV to 240 V.

- (a) What is the ratio of the number of secondary turns to the number of primary turns?
- (b) If the input current to the transformer is 2.0 A, what is the output current?
- (c) Determine the power loss in the transmission line if the total resistance of the transmission line is 200Ω .
- (d) What would the power loss have been if the transmission line was at 240 V the entire length of the line, rather than providing voltage at 12 kV? What does this say about transmission lines?

EXAMPLE

A transformer steps the line voltage down from 110 to 9.0 V so that a current of 0.50 A can be delivered to a doorbell.

- (a) What is the ratio of the number of turns in the primary and secondary windings?
- (b) What is the current in the primary winding?
- (c) What is the resistance seen by the 110-V source?

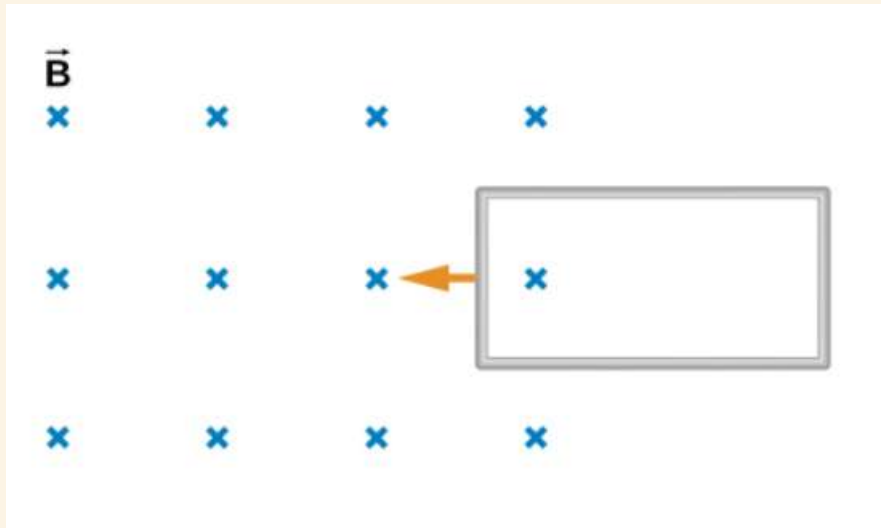
ANSWER:

(a) 12:1

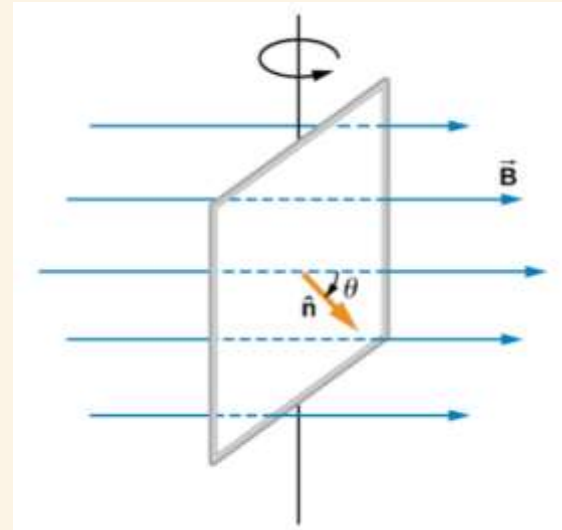
(b) 0.042 A

(c) 2.6×10^3

MOTIONAL EMF

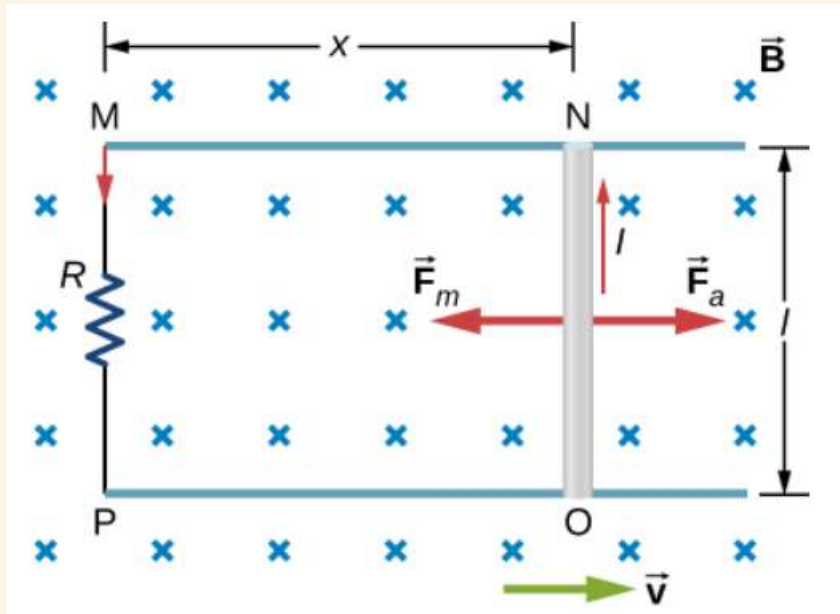


Magnetic flux changes as a loop moves into a magnetic field.



Magnetic flux changes as a loop rotates in a magnetic field.

MOTIONAL EMF



A conducting rod is pushed to the right at constant velocity. The resulting change in the magnetic flux induces a current in the circuit.

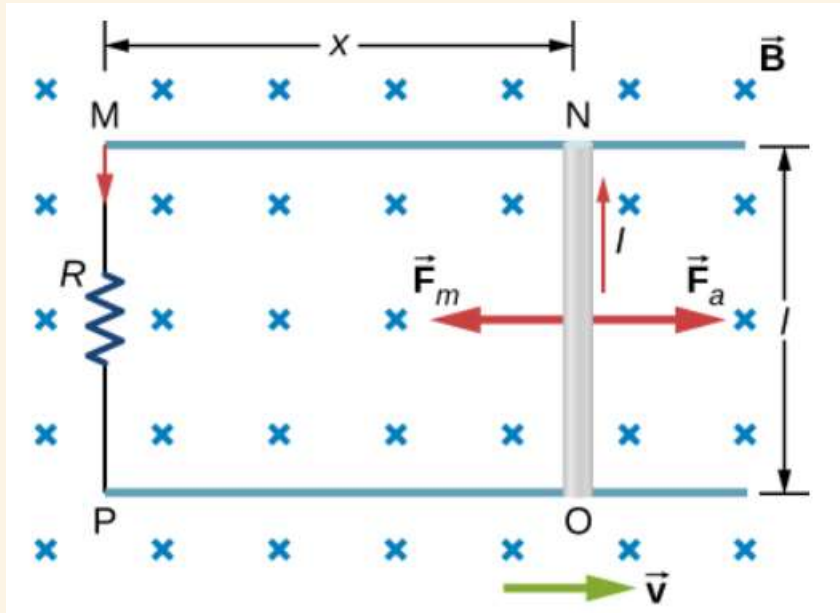
$$\Phi_m = \int_S \vec{B} \cdot \hat{n} dA$$

$$\Phi_m = Blx$$

The magnitude of the emf in terms of the moving conducting rod:

$$\epsilon = \frac{d\Phi_m}{dt} = Bl \frac{dx}{dt} = Blv$$

MOTIONAL EMF

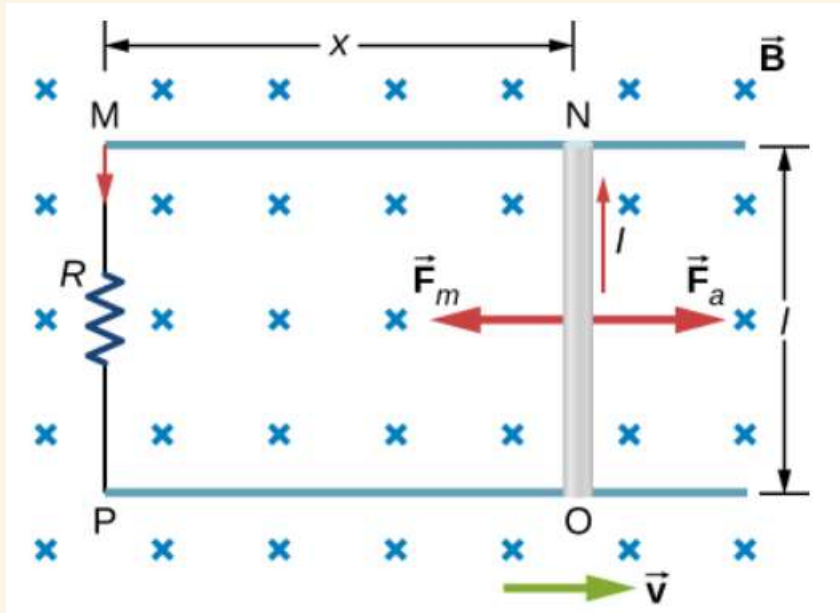


The current induced in the circuit is the emf divided by the resistance is:

$$I = \frac{Blv}{R}$$

The rod is moving at constant velocity, thus the applied force F_a must balance magnet force, $F_m = IlB$.

MOTIONAL EMF



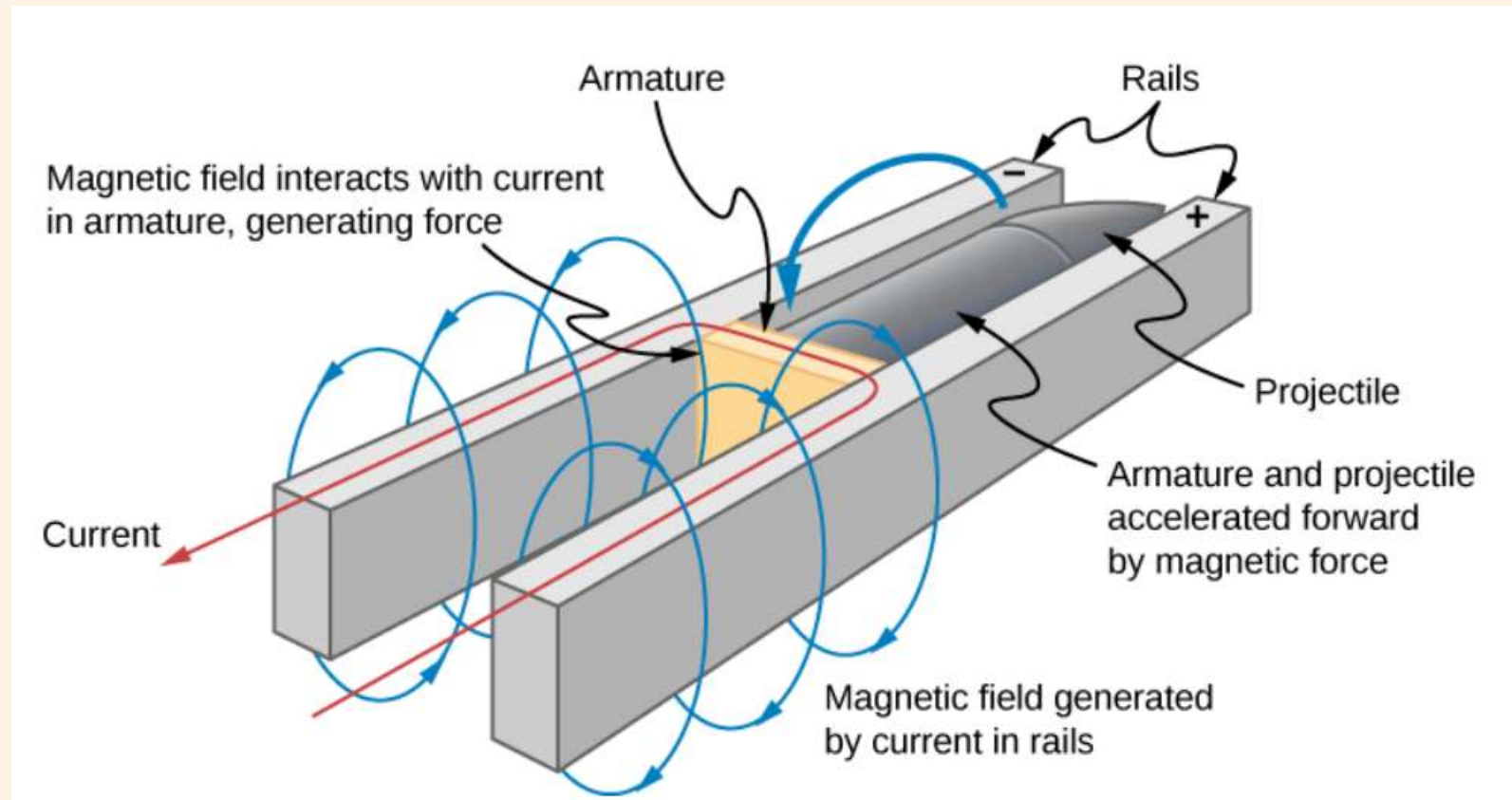
Thus, the power produced is:

$$F_a v = I l B v = \frac{B l v}{R} \cdot l B v = \frac{l^2 B^2 v^2}{R}$$

The power dissipated is:

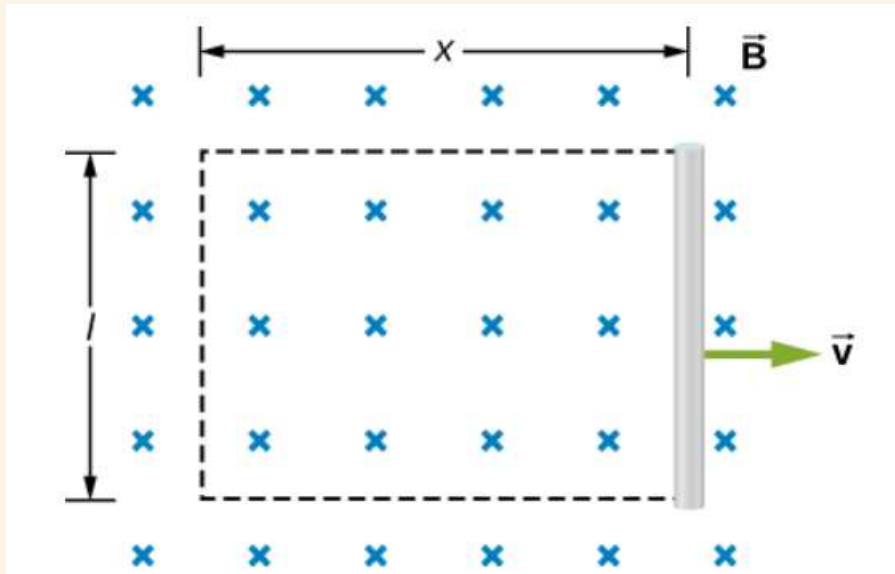
$$P = I^2 R = \left(\frac{B l v}{R} \right)^2 R = \frac{l^2 B^2 v^2}{R}$$

MOTIONAL EMF



Current through two rails drives a conductive projectile forward by the magnetic force created

MOTIONAL EMF



Let the moving rod be one side of the imaginary rectangular area. Thus, the magnetic flux through it is:

$$\Phi_m = Blx$$

By differentiating, emf is:

$$\epsilon = \frac{d\Phi_m}{dt} = Bl \frac{dx}{dt} = Blv$$

EXAMPLE

Calculate the motional emf induced along a 20.0-km conductor moving at an orbital speed of 7.80 km/s perpendicular to Earth's $5.00 \times 10^{-5} T$ magnetic field.

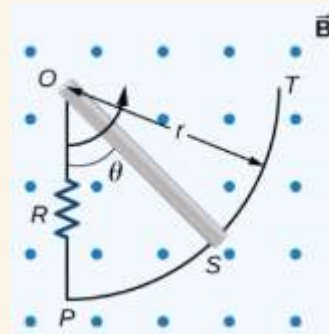
ANSWER:

$$7.8 \times 10^3 V$$

EXAMPLE

Figure below shows a metal rod **OS** that is rotating in a horizontal plane around point **O**. The rod slides along a wire that forms a circular arc **PST** of radius r . The system is in a constant magnetic field B that is directed out of the page.

- (a) If you rotate the rod at a constant angular velocity ω , what is the current I in the closed loop **OPSO**? Assume that the resistor R furnishes all of the resistance in the closed loop.
- (b) Calculate the work per unit time that you do while rotating the rod and show that it is equal to the power dissipated in the resistor.



SOLUTION

1. Find magnetic flux: $\Phi_m = BA$, where $A = \frac{r^2\theta}{2}$, therefore;

$$\Phi_m = B \frac{r^2\theta}{2}.$$

2. Find induced emf, by using $\omega = d\theta/dt$: $\epsilon = \frac{d\Phi_m}{dt}$, where $\Phi_m = B \frac{r^2\theta}{2}$, therefore;

$$\epsilon = \frac{Br^2\omega}{2}$$

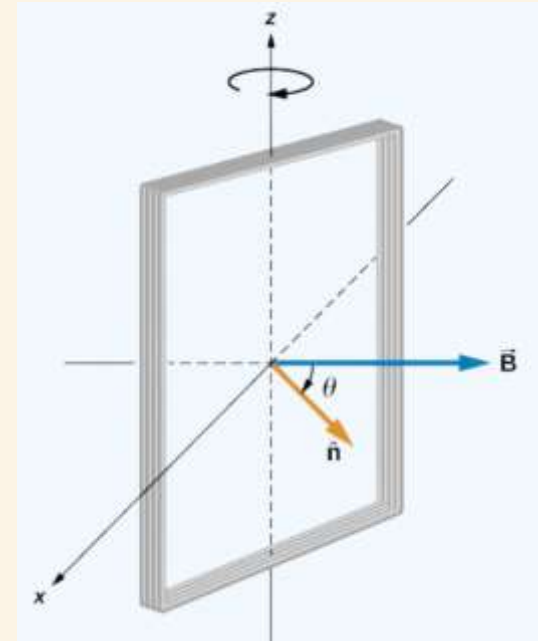
3. Substitute all known values in ϵ .

4. Find induced current (using Ohm's Law): $V = IR$, therefore;

$$I = \frac{Br^2\omega}{2}$$

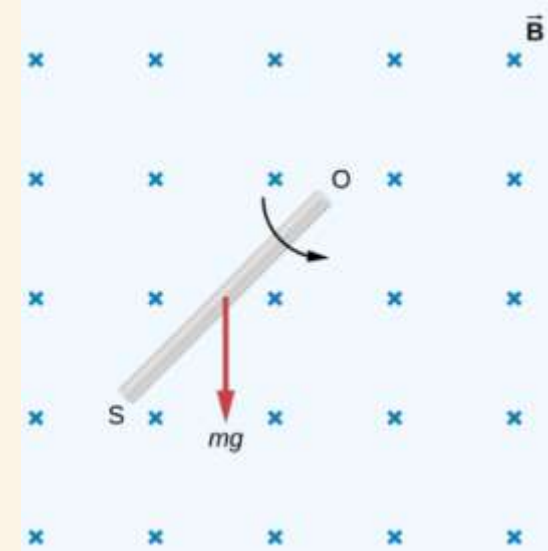
EXAMPLE

A rectangular coil of area $\mathbf{A}$ and $\mathbf{N}$ turns is placed in a uniform magnetic field $\mathbf{B} = B\mathbf{j}$, as shown in Figure. The coil is rotated about the $\mathbf{z}$ -axis through its center at a constant angular velocity ω . Obtain an expression for the induced emf in the coil.



EXAMPLE

Shown below is a rod of length l that is rotated counterclockwise around the axis through O by the torque due to mg . Assuming that the rod is in a uniform magnetic field B , what is the emf induced between the ends of the rod when its angular velocity is ω ? Which end of the rod is at a higher potential?



EXAMPLE

A rod of length 10 cm moves at a speed of 10 m/s perpendicularly through a 1.5-T magnetic field. What is the potential difference between the ends of the rod?

ANSWER:

1.5V



THANK YOU

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